

```
sent_tokenize(string1)

['In today's data science world Natural language processing has become the most popular topic of research.',
 'With Generative AI, ChatGPT, we are moving more and more towards NLP.',
 'So many new models like transformer models, Zero shot learning and more are coming up everyday.',
 'Therefore, with the world updating so fast, we also must make ourselves aware about these technologies.',
 'In this article we will focus on the basics of natural language processing and the application of these topics using one of the most popular NLP libraries known as NLTK - Natural Language Toolkit.']
```

Figure 2: Sentence tokeniser

Let us look at how we can do this using NLTK. First, import the required functions as shown below:

```
from nltk.tokenize import sent_tokenize, word_tokenize
```

I've given a sample string, which is the first paragraph of this article, as a sample for us to analyse. We are going to use the sentence tokeniser as shown below and in Figure 2.

```
sent_tokenize(string1)
```

It can be seen that the function returns a list of sentences in the given text. Now let us use the word tokeniser, as shown in Figure 3. As we can see, a list of words is returned here, and the tokens are on the word level.

```
word_tokenize(string1)
```

The next type of tokeniser is known as the punctuation tokeniser. We can use that with the following lines of code. We create the tokens based on the punctuation we give, as shown in Figure 4.

```
tokenizer = nltk.tokenize.PunktSentenceTokenizer()
sentences = tokenizer.tokenize(string1)
sentences
```

Stemming

The next major NLP process we are going to learn is called stemming. This is also a very important process in the field of NLP, in which we reduce a given word to its root form. For example, the words work, working and worked have the same sentiment or meaning. Therefore, it becomes easy for us

```
{x} 0s word_tokenize(string1)

['In',
 'today',
 ',',
 's',
 'data',
 'science',
 'world',
 'Natural',
 'language',
 'processing',
 'has',
 'become',
 'the',
 'most',
 'popular',
 'topic',
 'of',
 'research',
 '.',
 'With',
 'Generative',
 'AI',
```

Figure 3: Word tokeniser

to manage them, whether to create a vocabulary or for any other purpose. Stemming helps us improve the accuracy and the efficiency of the text. It basically ensures that all words with a similar meaning are treated uniformly.

There are multiple methods or algorithms that help us perform this task, the most popular being Porter Stemmer.

This method is defined by a set of rules that help us strip suffixes to obtain the base words, which are also known as stems. It's a useful preprocessing step in many NLP tasks like text classification, information retrieval, and sentiment analysis. We first import the stemmer and then initialise the Porter Stemmer as shown in Figure 5.

Let us now use the list of words we created above for checking out the stemming process. I've stored the list in the variable name 'l', using the following lines of code. We can call the stemmer on our list and print out the results.

```
after_stem = [s.stem(i) for i in l]
after_stem
```

```
tokenizer = nltk.tokenize.PunktSentenceTokenizer()
sentences = tokenizer.tokenize(string1)
sentences

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```

Figure 4: Punctuation tokeniser

```
[8] from nltk.stem import PorterStemmer
s = PorterStemmer()
```

Figure 5: Porter Stemmer